



A

Project Report

on

**UrbanEye: Satellite-Based Human Settlement
Detection System**

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “UrbanEye: Satellite-Based Human Settlement Detection System” which is submitted by Shreya Agarwal, Sameeksha Gupta and Shraddha Kushwaha in partial fulfilment of the requirement for the award of degree B. Tech. in Department of CSE(AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

Recent advances in satellite and aerial imaging technologies have significantly strengthened the role of remote sensing in applications such as urban planning, environmental monitoring, precision agriculture, and disaster management. High-resolution imagery enables detailed analysis of land use patterns, infrastructure development, and environmental changes. However, object detection in such images remains a challenging task due to complex scene structures, variations in object scale, occlusions, and visual similarities among object classes. Additionally, the limited availability of large, well-annotated datasets further affects model performance and generalization.

This project presents an enhanced object detection framework based on the YOLOv11 deep learning model, designed to achieve high detection accuracy and robust object localization in complex and dynamic environments. While earlier models such as Faster R-CNN and YOLOv5–YOLOv8 have demonstrated strong performance, the proposed approach introduces architectural improvements, attention mechanisms, and multi-scale feature extraction to better capture spatial and contextual information. Furthermore, dynamic data augmentation techniques, including rotation, scaling, flipping, and brightness adjustment, are applied to increase dataset diversity and reduce overfitting.

The model was trained and evaluated on a curated dataset consisting of diverse urban and natural scenes. Performance was measured using standard evaluation metrics such as precision, recall, F1-score, and mean Average Precision (mAP). Experimental results indicate that the proposed YOLOv11-based framework outperforms existing approaches in terms of accuracy, robustness, and computational efficiency, particularly in densely populated and heterogeneous environments.

Overall, the proposed system provides a scalable, efficient, and reliable solution for real-world remote sensing applications, supporting improved decision-making in urban development, environmental sustainability, and disaster response management.

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LIST OF ABBREVIATIONS

CNN	Convolution Neural Network
R-CNN	Region Convolution Neural Network
YOLO	You Only Look Once
CNN	Convolution Neural Network
HOG	Histogram of Oriented Gradients
SVM	Support Vector Machine
RPN	Region Proposal Network
SSD	Single Shot MultiBox Detector
mAP	Mean Average Precision
SPP	Spatial Pyramid Pooling
TP	True Positive
FN	False Negative
FP	False Positive
GIS	Geographic Information System

CHAPTER 1

INTRODUCTION

1.1 Introduction

1.1.1 Background

Remote sensing has emerged as one of the most powerful technologies for observing and analyzing the Earth's surface. With the rapid advancement of satellite systems, aerial imaging platforms, and geospatial technologies, large volumes of high-resolution imagery are continuously being generated. These images provide detailed information about land cover, infrastructure, vegetation, water bodies, and other environmental features. As a result, remote sensing has become an essential tool for applications such as environmental monitoring, urban planning, agricultural assessment, disaster management, and infrastructure development.

Traditional remote sensing analysis relied heavily on manual interpretation and classical image processing techniques. Experts would analyze satellite images to identify objects and patterns based on visual inspection or predefined rules. Although this approach provided valuable insights, it was time-consuming, labor-intensive, and often subject to human bias. Furthermore, the exponential increase in satellite imagery data has made manual analysis increasingly impractical.

To address these challenges, automated image analysis techniques have been developed to extract meaningful information from remote sensing imagery. Early automated approaches used handcrafted features such as texture descriptors, edge detection, and spectral indices combined with machine learning algorithms. These methods helped improve efficiency but often struggled to generalize across different geographic regions and imaging conditions.

The recent emergence of deep learning has revolutionized the field of computer vision and significantly enhanced the capabilities of remote sensing image analysis. Deep neural networks, particularly convolutional neural networks (CNNs), have demonstrated remarkable success in tasks such as image classification, segmentation, and object detection. By automatically learning hierarchical feature representations

from raw data, deep learning models are able to capture complex spatial patterns and relationships that traditional approaches fail to detect.

1.1.2 Importance of the Problem

Accurate detection of objects and land-cover elements in remote sensing imagery is critical for addressing several global challenges. Rapid urbanization, climate change, environmental degradation, and natural disasters require continuous monitoring and efficient management of natural and built environments. Automated object detection systems can help identify key features such as buildings, roads, vegetation, forests, and water bodies, enabling better planning and decision-making.

For example, urban planners can use object detection techniques to analyze infrastructure growth and land-use patterns in rapidly expanding cities. Environmental scientists rely on remote sensing data to monitor deforestation, habitat loss, and vegetation health. Similarly, disaster management authorities use satellite imagery to assess damage caused by floods, earthquakes, or landslides and to plan relief operations.

However, detecting objects in remote sensing imagery is significantly more challenging than detecting objects in natural images. Satellite images often contain extremely large spatial dimensions, complex backgrounds, and objects that appear at multiple scales and orientations. Additionally, objects may be densely distributed, partially occluded, or visually similar to surrounding features. These characteristics make accurate detection and classification a complex task that requires advanced computational techniques.

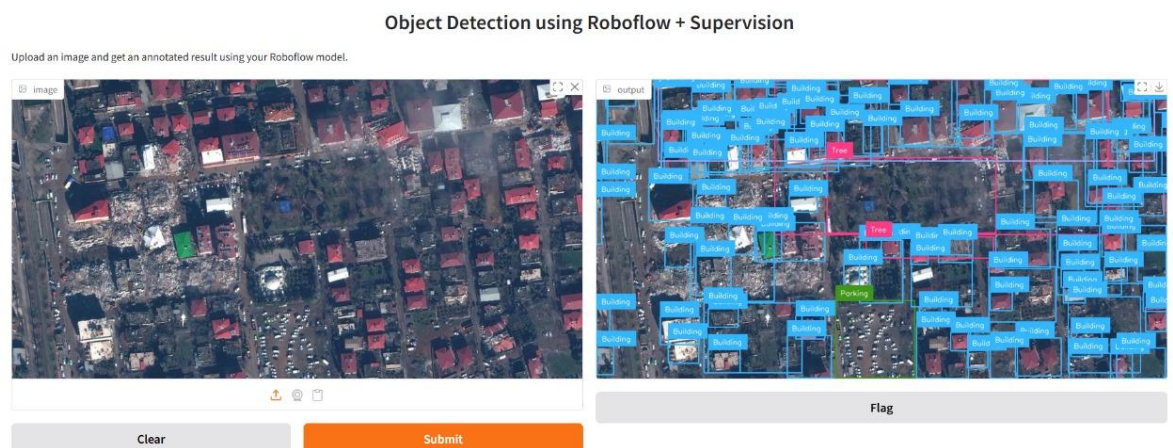


Fig 1.1: Objection Detection using satellite image

1.1.3 Current Trends

Recent research in remote sensing image analysis has increasingly focused on deep learning-based object detection frameworks. Models such as Faster R-CNN, RetinaNet, and the YOLO (You Only Look Once) family have achieved state-of-the-art performance in computer vision tasks. These architectures utilize advanced mechanisms including convolutional feature extractors, feature pyramid networks, anchor boxes, and attention modules to detect objects with high accuracy.

Among these approaches, the YOLO family of models has gained significant attention due to its ability to perform real-time object detection while maintaining competitive accuracy. Versions such as YOLOv5, YOLOv7, and YOLOv8 have been widely applied in various domains including surveillance, autonomous driving, medical imaging, and remote sensing.

Despite these advancements, applying these models directly to remote sensing imagery still presents several challenges. Many existing models struggle to detect small objects such as vehicles or small buildings in large satellite images. Additionally, overlapping objects and high object density often reduce detection accuracy. Variations in spatial resolution, illumination, and environmental conditions further complicate the detection process.

To overcome these limitations, recent research has focused on improving feature extraction mechanisms, enhancing multi-scale feature representation, and introducing attention-based architectures. These developments aim to improve the model's ability to capture complex spatial patterns and detect objects more accurately in challenging environments.

1.1.4 Motivation for the Project

Given the increasing availability of high-resolution satellite imagery and the growing demand for automated geospatial analysis, there is a strong need for robust object detection frameworks that can effectively handle the complexities of remote sensing data. Existing detection models often exhibit limitations when dealing with small objects, densely packed targets, and diverse environmental conditions commonly found in satellite imagery.

Recent advancements in the YOLO architecture, particularly the introduction of YOLOv11, provide new opportunities to address these challenges. YOLOv11 incorporates architectural improvements such as adaptive anchor-free detection mechanisms, improved feature fusion strategies, and cross-scale attention modules. These enhancements enable the model to capture richer spatial information and improve detection accuracy across objects of varying sizes.

Motivated by these developments, our project aims to explore the application of YOLOv11 for multi-class object detection in high-resolution remote sensing imagery. Unlike many previous studies that focus on detecting a single category of objects, our research aims to build a system capable of identifying multiple urban and environmental features simultaneously.

By combining a carefully curated dataset, data augmentation techniques, and transfer learning strategies, the proposed framework seeks to improve detection accuracy and model generalization across diverse geographic environments. The outcomes of this research can contribute to more efficient geospatial analysis systems that support urban planning, environmental monitoring, and disaster management.

1.2 Project Description

1.2.1 Problem Statement

Remote sensing imagery provides valuable spatial information for monitoring and analyzing natural and human-made environments. However, extracting meaningful insights from large volumes of satellite data remains a challenging task. Manual interpretation of imagery is time-consuming, inefficient, and difficult to scale for large geographic regions.

Automated object detection using deep learning has shown promising results, but many existing models face limitations when applied to remote sensing imagery. These challenges include difficulty detecting small objects, handling overlapping or densely distributed targets, and adapting to variations in spatial resolution and environmental conditions.

Therefore, there is a need for a robust and scalable object detection framework capable of accurately identifying multiple land-cover and infrastructure elements in high-resolution satellite imagery.

1.2.2 Objectives of the Project

The main objective of this project is to develop an efficient object detection framework using YOLOv11 for analyzing remote sensing imagery. The specific objectives include:

- To design and implement a multi-class object detection model using YOLOv11.
- To create and preprocess a dataset containing various urban and environmental features such as buildings, roads, vegetation, forests, and water bodies.
- To apply data augmentation techniques to improve dataset diversity and model robustness.
- To utilize transfer learning to enhance the model's performance and generalization capability.
- To evaluate the model using standard performance metrics such as precision, recall, and mean Average Precision (mAP).

1.2.3 Scope of the Project

This project focuses on the detection of multiple objects in high-resolution remote sensing images using deep learning techniques. The study primarily involves dataset preparation, model training, performance evaluation, and analysis of results.

The proposed framework aims to detect several categories of land-cover and infrastructure elements commonly found in satellite imagery. The system will be trained using a curated dataset and evaluated on its ability to accurately identify objects under varying conditions.

While the project demonstrates the feasibility of applying YOLOv11 for remote sensing analysis, it focuses mainly on the development and evaluation of the detection pipeline rather than the deployment of a large-scale operational system.

The proposed object detection framework has several practical applications across multiple domains like Urban Planning, Environmental Monitoring, Disaster Management and Geospatial Analysis.

CHAPTER 2

LITERATURE REVIEW

2.1 Traditional and Existing Object Detection Methods

Early research in object detection primarily relied on hand-crafted features and classical machine learning classifiers. These approaches required extensive domain knowledge and manual feature engineering, which limited their adaptability to complex real-world environments.

Krishna et al. (2012) proposed an object detection framework based on Haar-like features combined with the AdaBoost classifier. The method demonstrated reasonable performance for simple scenes with controlled lighting conditions. However, its dependency on manually designed features resulted in poor generalization when applied to complex urban environments involving occlusion, scale variation, and illumination changes.

Similarly, Wang et al. (2019) employed Histogram of Oriented Gradients (HOG) features with a Support Vector Machine (SVM) classifier for object recognition. While HOG-SVM-based methods improved robustness against minor variations, their detection speed was slow, and performance degraded significantly in high-density urban scenes.

Tu et al. (2018) explored Bag-of-Words (BoW) models for object classification. Although BoW-based approaches offered improved feature representation, they failed to capture spatial relationships between objects, which is crucial in urban scene analysis.

Overall, traditional approaches suffer from:

- Heavy reliance on manual feature extraction
- Poor performance in cluttered scenes
- Inability to scale for real-time applications

These limitations motivated the transition toward deep learning-based object detection techniques.

2.2 Deep Learning-Based Object Detection Approaches

The rapid advancement of Convolutional Neural Networks (CNNs) revolutionized object detection by enabling automatic extraction of high-level semantic features directly from images.

Deep learning-based object detection algorithms are broadly classified into:

- Two-stage detectors
- One-stage detectors

2.2.1 Two-Stage Detection Algorithms

Girshick et al. (2014) introduced R-CNN, which first generates region proposals and then performs classification and bounding box regression. Although R-CNN achieved high detection accuracy, it was computationally expensive and unsuitable for real-time applications.

Fast R-CNN (Girshick et al., 2015) improved efficiency by sharing convolutional computations across proposals. Further enhancement was achieved by Faster R-CNN (Ren et al., 2017), which integrated a Region Proposal Network (RPN). Despite improved accuracy, two-stage detectors still suffer from high inference time and complex architectures.

Two-stage methods are best suited for applications where accuracy is prioritized over speed, such as medical imaging and offline analysis.

2.2.2 One-Stage Detection Algorithms

To overcome the speed limitations of two-stage detectors, one-stage detection algorithms were introduced.

Liu et al. (2016) proposed Single Shot MultiBox Detector (SSD), which performs object localization and classification in a single forward pass. SSD achieved faster detection but struggled with small object detection.

Redmon et al. (2016, 2017, 2018) introduced the YOLO series, which reformulated object detection as a regression problem. YOLO-based models demonstrated significantly higher frame rates compared to Faster R-CNN, making them suitable for real-time applications.

Bochkovskiy et al. (2020) further improved YOLO by introducing YOLOv4, incorporating features such as CSPDarknet, PANet, and advanced data augmentation techniques.

2.3 Evolution of YOLO-Based Object Detection Models

Object detection has evolved significantly with the introduction of the YOLO (You Only Look Once) family of models, which reformulate object detection as a single-stage regression problem. Unlike two-stage detectors such as R-CNN and Faster R-CNN, YOLO models predict bounding boxes and class probabilities directly from the input image in a single forward pass, enabling real-time detection [1], [10].

2.3.1 YOLOv1 and YOLOv2

Redmon et al. introduced YOLOv1 in 2016 as the first unified real-time object detection framework [1]. YOLOv1 divided the input image into grid cells and predicted bounding boxes and class probabilities simultaneously. While YOLOv1 achieved remarkable speed improvements over traditional detectors, it suffered from localization errors and poor performance in detecting small objects.

To address these limitations, YOLOv2 (YOLO9000) was proposed by Redmon and Farhadi in 2017 [2]. YOLOv2 introduced:

- Anchor boxes
- Batch normalization
- Higher-resolution input images

These improvements significantly enhanced detection accuracy while maintaining real-time performance. However, challenges in detecting small and densely packed objects remained, particularly in aerial and urban imagery [2].

2.3.2 YOLOv3

Redmon and Farhadi proposed YOLOv3 in 2018, incorporating multi-scale detection using feature maps at different resolutions [3]. YOLOv3 employed the Darknet-53 backbone network, which improved feature extraction capability.

From the reviewed literature, YOLOv3 demonstrated:

- Improved small object detection compared to earlier YOLO versions
- Better class prediction using logistic classifiers

However, YOLOv3 increased model complexity and computational cost, making it less suitable for lightweight or low-resource deployment scenarios [3], [13].

2.3.3 YOLOv4

Bochkovskiy et al. introduced YOLOv4 in 2020, focusing on optimizing speed and accuracy through architectural and training improvements [4]. YOLOv4 integrated:

- CSPDarknet backbone
- PANet and Spatial Pyramid Pooling (SPP) modules
- Advanced data augmentation techniques

YOLOv4 achieved state-of-the-art performance among one-stage detectors while maintaining real-time inference speed. Despite its strong performance, YOLOv4 remained computationally intensive for embedded and edge-device environments [4], [13].

2.3.4 YOLOv5 and Its Variants

YOLOv5 marked a significant transition by being implemented in the PyTorch framework, improving flexibility, training efficiency, and deployment [5]. It introduced multiple variants (YOLOv5s, YOLOv5m, YOLOv5l, YOLOv5x) to balance speed and accuracy.

Wang et al. [6] applied YOLOv5s for multi-class object detection in urban scenes using high-resolution remote sensing images. Their study focused on detecting buildings, vehicles, and trees from the ISPRS Potsdam dataset.

The experimental results demonstrated that YOLOv5s achieved:

- Mean Average Precision (mAP) of 82.83%
- Detection speed of 44.14 FPS
- Better overall performance compared to SSD and Faster R-CNN

Although YOLOv5s showed slightly lower F1-scores for building and vehicle detection compared to Faster R-CNN, its superior speed and lightweight architecture made it ideal for real-time urban applications [6].

Several other studies confirmed the effectiveness of YOLOv5-based models in applications such as crack detection, vessel detection, and agricultural monitoring, highlighting their robustness and deployment efficiency [25], [26], [27].

2.3.5 YOLOv6 and YOLOv7

Subsequent improvements were introduced with YOLOv6 and YOLOv7, focusing on industrial deployment and performance optimization [15], [14].

Comparative experiments conducted on aerial datasets such as VisDrone2019 revealed that:

- YOLOv6 achieved marginal accuracy improvements but at the cost of significantly increased model size
- YOLOv7 introduced *trainable bag-of-freebies* techniques, achieving higher accuracy while maintaining real-time performance

However, these models require greater computational resources, limiting their applicability in lightweight urban monitoring systems [14], [15].

2.3.6 YOLOv8

YOLOv8 represents the latest generation of YOLO models and introduces architectural refinements aimed at improving small object detection performance. According to recent state-of-the-art surveys on remote sensing object detection, YOLOv8 demonstrated:

- Detection accuracy exceeding 90% for certain object classes
- Inference time as low as 0.2 ms in optimized environments
- Improved generalization through transfer learning techniques

Despite these advantages, YOLOv8 models are computationally heavier compared to YOLOv5s, which may restrict deployment on edge devices and real-time urban systems [13], [16].

2.4 Comparative Analysis of YOLO Models (v1–v8)

To clearly understand the progression and suitability of different YOLO versions for object detection tasks, especially in urban and aerial remote sensing environments, a comparative analysis is presented based on architecture, performance, advantages, and limitations reported in the reviewed literature.

Table 2.1: Comparison of YOLO Models Used in Literature

YOLO Version	Year	Key Architectural Features	Advantages	Limitations	Major Findings from Literature
YOLOv1[1]	2016	Single CNN, grid-based prediction	Very fast, end-to-end detection	Poor localization, weak small object detection	Introduced real-time detection but inaccurate for dense scenes
YOLOv2 (YOLO9000) [2]	2017	Anchor boxes, batch normalization	Improved accuracy over YOLOv1	Still struggles with small objects	Better trade-off between speed and accuracy
YOLOv3 [3]	2018	Darknet-53, multi-scale detection	Better small object detection	Increased computational cost	Improved detection in complex scenes paper1
YOLOv4 [4],[13]	2020	CSPDarknet, PAN, SPP	High accuracy and speed	Heavy for edge devices	Strong real-time performance paper1
YOLOv5 [5], [6]	2020–2022	PyTorch-based, lightweight variants	Fast, compact, deployable	Slight loss in accuracy for large objects	Best balance of speed & accuracy
YOLOv5s[5], [6]	2024	CSPDarknet53, FPN + PAN	mAP 82.83%, 44 FPS	Slightly lower F1 for buildings	Excellent for urban detection paper1
YOLOv6 [15]	2022	Industrial optimization	Improved accuracy	Larger model size	Limited gains over YOLOv5 paper 3

YOLOv7 [14]	2023	Trainable bag-of-freebies	High accuracy	Computationally heavy	Strong but resource intensive
YOLOv8 [13], [16]	2023	Anchor-free design, improved head	>90% accuracy, fast inference	Not lightweight	Best small-object performance

2.5 Reviewed Gap

Although the YOLO family of object detection models has evolved significantly from YOLOv1 to YOLOv8, the reviewed literature highlights several persistent challenges. Early YOLO versions (YOLOv1–YOLOv3) prioritized detection speed but exhibited poor localization accuracy and limited performance in complex scenes involving small or densely packed objects. Subsequent versions such as YOLOv4 and YOLOv5 introduced architectural optimizations that improved accuracy and inference speed; however, their performance remains constrained when dealing with high variability in object scale, occlusion, and background complexity.

More recent models, including YOLOv6, YOLOv7, and YOLOv8, have focused on improving detection accuracy—particularly for small objects—through advanced feature fusion strategies and architectural refinements. While these models demonstrate superior performance, the literature indicates that such improvements are often achieved at the cost of increased model complexity and higher computational requirements, limiting their suitability for real-time deployment on resource-constrained systems.

Thus, the existing literature reveals a gap in achieving an optimal balance between detection accuracy, inference speed, and computational efficiency, motivating the exploration of newer YOLO-based architectures that can effectively address these limitations.

2.6 YOLO Models in Urban and Aerial Image Detection

Urban and aerial imagery presents challenges such as small object size, dense object distribution, and occlusions. YOLOv5s has been widely adopted in such scenarios due to its lightweight architecture and fast inference speed.

Wang et al. (2024) demonstrated that YOLOv5s achieved an mAP of 82.83% while maintaining real-time detection at 44.14 FPS, outperforming SSD and Faster R-CNN in overall performance [6]. In contrast, although YOLOv7 and YOLOv8 achieved higher accuracy, their increased parameter count and computational demand reduced suitability for resource-constrained urban monitoring systems [14], [16].

2.7 Small Object Detection and YOLO Enhancements

Small object detection remains a major challenge in aerial imagery. Studies involving YOLOv6, YOLOv7, and YOLOv8 introduced enhancements such as:

- Feature pyramid optimization
- Context aggregation modules
- Anchor-free detection heads

Recent improvements, including MPE-YOLO and ES-C2f modules, showed measurable gains in mean Average Precision for small targets, but at the cost of increased latency and model complexity [15], [16].

2.8 Performance Trade-off Analysis

The reviewed literature consistently highlights the trade-off between:

- Accuracy and inference speed
- Model size and deployment feasibility

YOLOv5s emerges as the most balanced model, offering competitive accuracy, high processing speed, and minimal memory footprint. This balance is critical for smart city applications, traffic monitoring, and real-time urban surveillance [6], [13].

2.9 Consolidated Research Gap

Based on the comprehensive review of YOLO-based object detection models, the following gaps are identified:

- Early YOLO models lack robustness in handling complex scenes with small, overlapping, or densely distributed objects.

- Advanced YOLO variants achieve higher detection accuracy but require substantial computational resources, limiting real-time and edge-device deployment.
- Existing studies primarily focus on improving accuracy, with comparatively less emphasis on achieving scalability and efficiency simultaneously.

Therefore, there exists a clear need for an improved object detection framework that builds upon the strengths of previous YOLO models while overcoming their limitations in terms of accuracy, speed, and computational efficiency. This motivates the adoption and evaluation of a newer YOLO-based approach in this project.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Architecture

The proposed UrbanEye system is designed to detect and analyze human settlements from aerial imagery using deep learning techniques. The system architecture consists of several interconnected stages that process high-resolution aerial images and generate meaningful analytical outputs.

The architecture begins with the image input stage where aerial or satellite images are collected. These images typically contain large geographic regions including buildings, vegetation, water bodies, and infrastructure. Because these images are extremely large in size, direct processing is computationally expensive.

To solve this issue, an image slicing process is applied where large images are divided into smaller tiles of 640×640 pixels. This allows the deep learning model to process the images efficiently while preserving local spatial details.

The processed images are then passed to the YOLOv11 detection model. The model uses deep convolutional neural networks to analyze the visual features present in each image tile and detect urban objects.

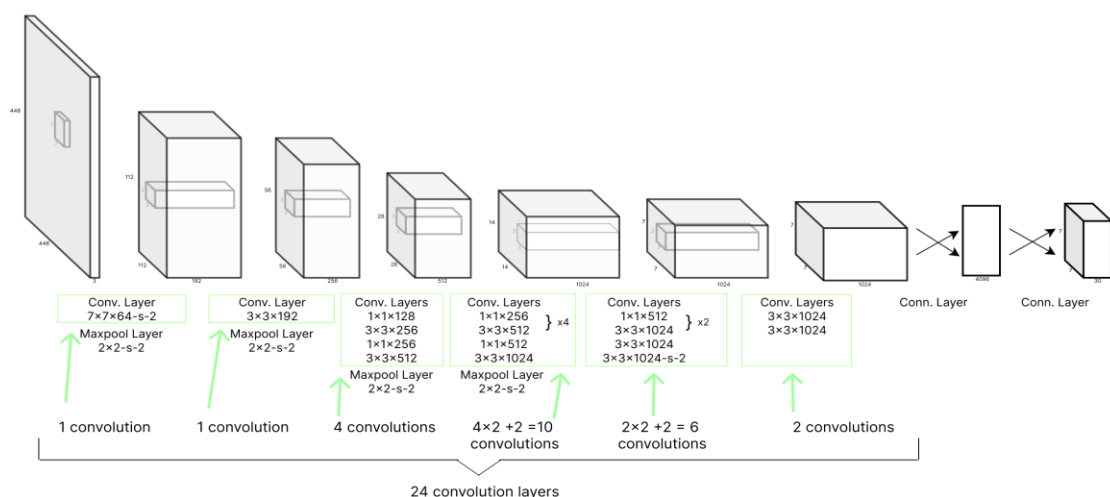


Fig 3.1: YOLO Architecture

After detection, the results are analyzed to identify different classes of objects such as buildings, vegetation, and water bodies. The detected outputs are further processed to generate statistical insights which help in understanding settlement patterns within the aerial images.

3.2 Workflow

The workflow of the UrbanEye system describes the sequence of operations from data collection to final analysis. The process begins with the collection of aerial and satellite imagery representing urban environments.

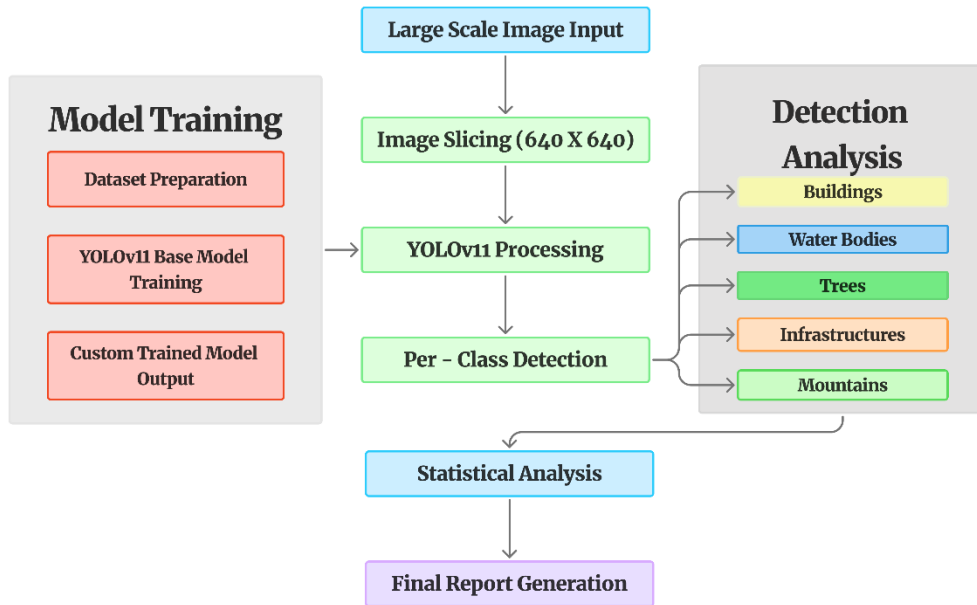


Fig 3.2: Flowchart for the urban object detection model development

Since aerial images are extremely large, they undergo preprocessing where the images are divided into smaller 640×640 -pixel tiles. This step improves computational efficiency and increases the number of training samples.

After preprocessing, the dataset is annotated by labeling different objects present in the images. These annotations help the model learn how to distinguish between various urban structures.

Once labeling is completed, the dataset is processed through the Roboflow platform. Roboflow automatically organizes the dataset, prepares it for training, and divides it into training, validation, and testing sets.

The prepared dataset is then used to train a YOLOv11 object detection model. After training, the model predicts bounding boxes and class labels for objects present in aerial images. The detection results are finally analyzed to evaluate the model performance and generate reports.

3.3 Algorithm

The detection algorithm used in this research is based on the YOLOv11 object detection framework. YOLO is a deep learning based real-time object detection algorithm that processes the entire image in a single pass through a convolutional neural network.

The algorithm first takes an input image and passes it through multiple convolutional layers that extract important features such as edges, textures, and shapes. These features help the model recognize different types of objects within the image.

The model divides the image into a grid and predicts bounding boxes and confidence scores for objects present in each grid cell. Each bounding box is associated with a class probability that determines the type of object detected.

The final output of the algorithm consists of bounding boxes around detected objects along with their predicted class labels. This allows the system to identify different urban features present in aerial images.

3.4 Tools and Technologies

Several tools and technologies were used during the development of the UrbanEye system.

- Roboflow was used for dataset management and annotation. It provides an interactive interface for labeling objects in images and preparing datasets for training.
- YOLOv11 architecture was used for object detection. The model was trained using the automated training pipeline provided within the Roboflow environment.

- Python and computer vision libraries were used for data processing and analysis.

These technologies collectively enabled efficient processing of aerial imagery and accurate detection of urban settlement features.

3.5 Data Collection

The dataset used in this research consists of high-resolution aerial and satellite images representing urban landscapes. These images contain a variety of environmental conditions and geographic locations to ensure that the model can generalize well to different scenarios.

The images include different types of urban features such as buildings, vegetation areas, water bodies, and infrastructure. High resolution images were selected to ensure that even small objects could be detected accurately.

Each large aerial image had a resolution of approximately 15000×15000 pixels. To make them suitable for model training, the images were divided into smaller 640×640 -pixel tiles.

After slicing, the dataset contained more than 6500 image samples which were used for training and evaluation purposes.

3.6 Implementation Steps

The implementation of the UrbanEye human settlement detection system involved a structured computer vision pipeline including dataset preparation, annotation, model training, and detection analysis.

- Initially, high-resolution aerial images were collected and prepared as a dataset. Because these images were extremely large, they were divided into smaller 640×640 -pixel tiles to make them compatible with deep learning models.
- The next step involved data annotation. The images were manually labeled using the Roboflow platform where bounding boxes were drawn around objects such as buildings, vegetation areas, and water bodies. Each labeled object was assigned a corresponding class category.

- After the labeling process was completed, the dataset was processed through Roboflow. The platform automatically prepared the dataset and divided it into training, validation, and testing subsets.
- The YOLOv11 model available within Roboflow was then used for training. The model learned to detect urban objects by analyzing patterns in the labeled images.
- After training, the model generated predictions in the form of bounding boxes and class labels for objects present in aerial images. The results were evaluated using analytical tools such as a confusion matrix to understand the detection performance and identify possible misclassifications.

This implementation demonstrates how annotated aerial imagery combined with automated training platforms like Roboflow can be used to effectively detect and analyze human settlements.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Result

The experimental results demonstrate that the proposed **YOLOv11-based object detection framework** performs effectively for detecting multiple urban and natural features from aerial imagery. The model achieved **77.4% precision, 58.0% recall, 60.3% mAP@50, and 54.7% mAP@50–95**, indicating strong detection capability across different intersection-over-union (IoU) thresholds.

The high precision value indicates that the model produces relatively few false positive detections, meaning that most predicted objects correspond to actual objects present in the images. Meanwhile, the recall score reflects the model’s ability to identify a large portion of the objects in the dataset. Although recall is slightly lower than precision, the overall balance between the two metrics demonstrates stable detection performance across diverse urban scenarios.

The mean Average Precision (mAP) values further confirm the model’s robustness. The obtained mAP@50 of 60.3% shows that the model performs well when the predicted bounding boxes closely overlap with ground-truth annotations. Moreover, the mAP@50–95 score of 54.7% indicates that the model maintains reliable detection performance even under stricter IoU thresholds, which require more accurate object localization.

Class-wise evaluation also revealed that the model performs exceptionally well for objects with distinct structural and geometric characteristics, such as airport runways, bridges, and trees. These objects achieved precision values exceeding 90%, primarily because their visual patterns are clearly distinguishable in aerial imagery. In contrast, road detection exhibited comparatively lower accuracy due to the presence of visually similar linear features such as rivers, shadows, and land boundaries.

Overall, the results demonstrate that the proposed framework effectively captures spatial features and object patterns in complex aerial scenes, making it suitable for applications such as urban mapping, infrastructure monitoring, and environmental analysis.

4.2 Performance Evaluation Metrics

The performance of the proposed model was evaluated using widely accepted object detection metrics, which provide comprehensive insight into both detection accuracy and classification reliability.

4.2.1 Mean Average Precision (mAP)

Mean Average Precision (mAP) is the primary evaluation metric used in this study. It represents the average detection accuracy across all object classes and is defined as:

$$mAP = \frac{1}{n} \sum_{k=1}^n AP(k)$$

Where:

- n is the number of classes
- $AP(k)$ is the average precision of class k

4.2.2 Precision

Precision measures how accurately the model predicts positive detections:

$$Precision = \frac{TP}{TP + FP}$$

Where:

- TP = True Positives
- FP = False Positives

4.2.3 Recall

Recall evaluates the model's ability to detect all relevant objects:

$$Recall = \frac{TP}{TP + FN}$$

Where:

- FN = False Negatives

4.2.4 F1 Score

The F1 score provides a balance between precision and recall:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

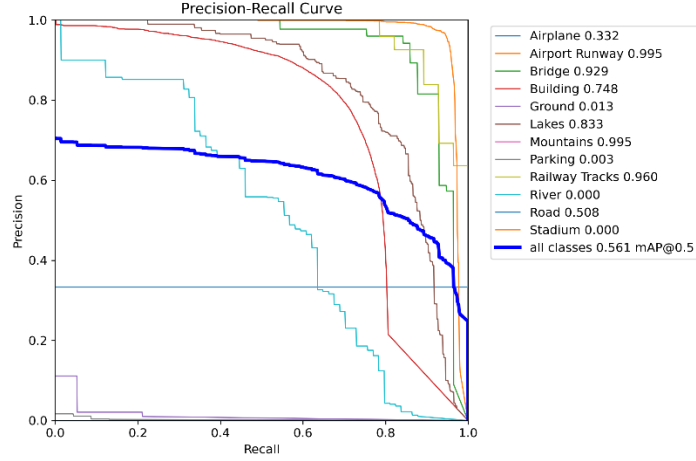


Fig 4.1: Performance evaluation curve of YOLOv11 on the proposed dataset

4.3 Experimental Analysis and Model Comparison

To assess the effectiveness of YOLOv11, its performance was compared with several state-of-the-art object detection models using the proposed dataset. The evaluation metrics included precision, recall, mAP@50, mAP@50–95, and inference speed.

Table 4.1: Model Performance Comparison

Method	Precision (%)	Recall (%)	mAP50 (%)	mAP50–95 (%)	Speed (ms)
Detectron2	50.0	32.7	16.0	24.0	0.9
YOLOv5	53.4	49.7	27.0	18.4	0.5
YOLOv6	53.2	47.4	32.1	16.6	0.4
YOLOv7	54.5	46.2	34.1	25.0	0.3
YOLOv8	68.0	60.0	43.0	17.5	0.2
YOLOv11	77.4	58.0	60.3	54.7	0.2

YOLOv11 outperformed all previous YOLO versions and Detectron2 in terms of precision and mAP. Architectural enhancements such as C3k2 **blocks** and C2PSA modules contributed to improved feature extraction and reduced computational

complexity. Despite achieving comparable recall, Detectron2 suffered from slower inference speeds, making YOLOv11 more suitable for real-time applications.

4.4 Experimental Analysis

The proposed dataset was employed to conduct a comprehensive evaluation of object detection performance using standard metrics, including precision, recall, mean Average Precision at 0.5 IoU (mAP@50), and mean Average Precision across multiple IoU thresholds (mAP@50–95). The comparative results, summarized in Table I, clearly demonstrate the superior performance of the YOLOv11 model over earlier YOLO variants and other state-of-the-art object detection frameworks.

Among all evaluated models, YOLOv11 achieved the highest overall detection accuracy, recording 77.4% precision, 58.0% recall, 60.3% mAP@50, and 54.7% mAP@50–95. These results indicate a significant improvement over YOLOv8, which achieved 68% precision, 60% recall, 43% mAP@50, and 17.5% mAP@50–95, highlighting YOLOv11’s enhanced ability to maintain high detection accuracy across varying intersection-over-union thresholds. Despite the increased accuracy, YOLOv11 maintained a competitive inference speed of approximately 7.6 milliseconds per image, making it suitable for real-time applications.

When compared with earlier versions such as YOLOv5, YOLOv6, and YOLOv7, YOLOv11 exhibited an improvement of approximately 22% in mAP@50 and achieved 30% faster processing speeds. These gains can be attributed to key architectural advancements, including the integration of C3k2 blocks and C2PSA modules, which significantly enhance feature representation while reducing computational overhead. Such improvements enable YOLOv11 to efficiently detect both small and large objects in complex urban environments.

In comparison, Detectron2 demonstrated relatively lower performance, achieving 50.0% precision, 32.7% recall, and 16.0% mAP@50, while operating at a substantially slower inference speed of 23 ms per image. Although Detectron2 achieved comparable recall in certain cases, its higher computational cost limits its practicality for real-time urban object detection tasks.

Overall, the experimental results confirm that YOLOv11 offers an optimal balance between accuracy and computational efficiency, outperforming existing models in both detection performance and processing speed. These characteristics make YOLOv11 particularly well-suited for real-time urban mapping, aerial surveillance, and large-scale geospatial analysis applications.

4.5 Class-wise Detection Performance

The class-wise evaluation of the YOLOv11 model on the Roboflow aerial imagery dataset highlights its strong capability to accurately detect and classify diverse urban and geographic features. Among all object categories, airport runways achieved outstanding performance, recording 100% precision in both the validation and test datasets. This exceptional accuracy can be attributed to the clear geometric structure and high visual contrast of runways in aerial images. Tree detection also demonstrated excellent results, achieving 97% precision during validation and 98% precision during testing, indicating the model's robustness in recognizing natural features with consistent textures and shapes.

Similarly, bridge detection yielded high precision scores of 92% on the validation set and 91% on the test set, reflecting the effectiveness of YOLOv11 in identifying elongated and structurally distinct objects. Mountain and lake classes exhibited reliable detection performance as well, with precision values exceeding 80% across both evaluation phases, despite variations in terrain and background complexity. Building detection achieved 77% precision during validation and 79% precision during testing, demonstrating strong performance given the dense layouts, occlusions, and architectural diversity typically present in urban environments.

In contrast, road detection showed comparatively moderate precision, with 59% on the validation set and 49% on the test set. This reduction in accuracy is likely due to the visual similarity of roads with other linear features such as rivers, shadows, and pathways, as well as partial occlusions in dense urban regions. Overall, the class-wise results confirm that YOLOv11 is particularly effective in detecting objects with well-defined boundaries and distinctive visual patterns, while highlighting potential areas for improvement in classes with ambiguous or visually complex characteristics.

4.6 Visualization Analysis

The qualitative visualization results further validate the effectiveness of the proposed YOLOv11-based detection framework. Detection outputs, represented through bounding boxes and corresponding class labels, indicate stable and consistent performance across a wide range of urban and geographic features present in aerial imagery. The model demonstrated exceptionally high detection accuracy for airport runways, trees, and bridges, with precision values exceeding 90%. This strong performance highlights the model's ability to recognize objects with clear geometric structures, distinct textures, and well-defined boundaries.

Vegetation and buildings were also detected with high reliability, achieving precision values of approximately 85% and 79%, respectively. Notably, these results were maintained even in densely populated urban regions and complex natural environments, where object overlap and background variation are common challenges. The detection of mountains and lakes showed robust performance, with precision consistently above 80%, indicating that the model effectively captured large-scale geographic features despite changes in terrain, lighting, and background complexity.

In contrast, road detection achieved moderate accuracy. This limitation can be attributed to the visual similarity of roads with other linear features such as riverbanks, shadows, or land boundaries, which occasionally led to misclassification. Nevertheless, the overall visualization analysis confirms that YOLOv11 effectively detects both large infrastructure elements and natural features, particularly excelling in scenes with dense and clearly structured objects. These findings emphasize the model's suitability for real-world applications such as urban mapping, land-use analysis, and environmental monitoring, where robustness across diverse visual conditions is critical.

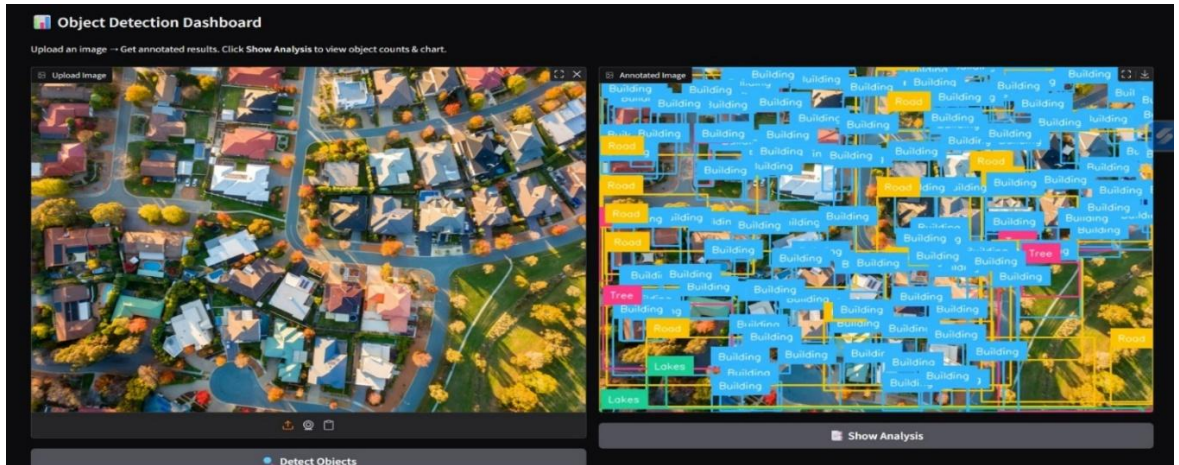


Fig 4.2: Object Detection represented through bounding boxes and corresponding class labels.

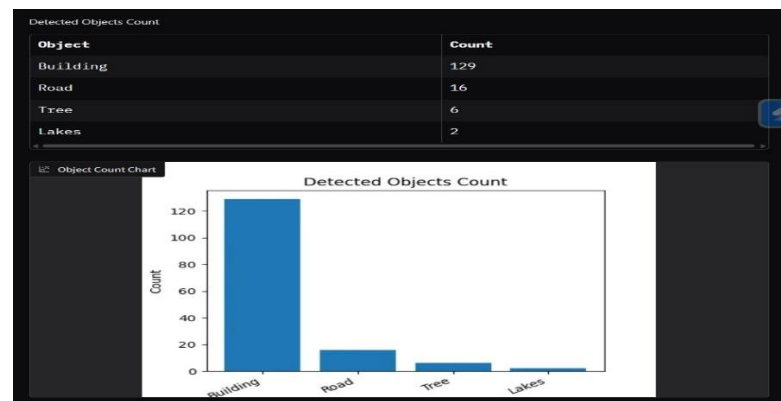


Fig 4.3: Detected Objects Count

4.7 Discussion

4.7.1 Comparison with Existing system

To evaluate the effectiveness of the proposed model, the performance of YOLOv11 was compared with several widely used object detection models, including Detectron2, YOLOv5, YOLOv6, YOLOv7, and YOLOv8.

The experimental comparison indicates that YOLOv11 consistently outperforms previous YOLO versions and the Detectron2 framework in terms of precision and mean Average Precision. For example, YOLOv8 achieved 68% precision and 43% mAP@50, whereas YOLOv11 improved these metrics to 77.4% precision and 60.3% mAP@50, demonstrating a substantial improvement in detection accuracy.

Earlier models such as YOLOv5, YOLOv6, and YOLOv7 achieved lower detection accuracy and struggled to maintain stable performance across varying IoU thresholds. These models also showed limitations when detecting objects with complex textures or irregular shapes in aerial imagery.

In contrast, YOLOv11 incorporates several architectural enhancements, including C3k2 blocks and C2PSA modules, which significantly improve feature extraction and spatial attention mechanisms. These improvements enable the model to better capture both fine-grained details and large-scale spatial patterns, resulting in higher detection accuracy for both small and large objects.

Another important advantage of YOLOv11 is its efficient inference speed. The model processes images in approximately 7.6 milliseconds per frame, which is significantly faster than the 23 milliseconds per frame required by Detectron2. This efficiency makes YOLOv11 highly suitable for real-time aerial monitoring and large-scale geospatial analysis systems.

Overall, the comparative evaluation confirms that YOLOv11 provides an optimal balance between accuracy, speed, and computational efficiency, making it a superior choice for aerial object detection tasks.

4.7.2 Strengths of the model

The proposed YOLOv11-based detection system offers several advantages that contribute to its strong performance in aerial image analysis.

First, the model achieves high detection accuracy, particularly for objects with well-defined shapes such as airport runways, bridges, and trees. The high precision values demonstrate the model's ability to correctly identify objects with minimal false detections.

Second, the advanced architecture of YOLOv11 enables improved feature representation and object localization. Components such as C3k2 blocks and C2PSA modules enhance the model's ability to capture complex spatial relationships within aerial imagery.

Third, the model demonstrates fast inference speed, making it suitable for real-time applications such as drone surveillance, smart city monitoring, and disaster management. The ability to process images in milliseconds allows the system to operate efficiently in large-scale datasets and continuous monitoring environments.

Another advantage is the scalability of the training pipeline, facilitated by the Roboflow platform. The platform simplifies data annotation, preprocessing, and dataset management, enabling efficient model development and experimentation.

Finally, the model exhibits strong generalization capability, successfully detecting multiple object categories across diverse urban and natural landscapes.

4.7.3 Weakness of the model

Despite its strong performance, the proposed system also presents several limitations that should be addressed in future work. One limitation is the moderate recall performance, which indicates that some objects in the dataset remain undetected. This issue is particularly noticeable for classes such as roads, where visual ambiguity and occlusion can affect detection accuracy. Another challenge is the difficulty in distinguishing visually similar features, such as roads, rivers, and shadows. These similarities occasionally lead to misclassification or missed detections. Additionally, the model performance is dependent on the quality and diversity of the training dataset. Limited variation in certain classes may restrict the model's ability to generalize to entirely new environments or unseen geographic regions. Finally, while the model performs efficiently on high-resolution aerial imagery, extremely dense urban scenes or severe occlusions may still impact detection accuracy.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

Our project focused on developing an efficient and robust object detection framework for analysing high-resolution remote sensing imagery using deep learning techniques. Remote sensing plays a crucial role in monitoring and managing natural and built environments, providing valuable spatial information for applications such as urban planning, environmental monitoring, disaster management, and infrastructure development. However, extracting meaningful information from large volumes of satellite imagery is a challenging task due to factors such as complex backgrounds, varying spatial resolutions, and the presence of objects at multiple scales. To address these challenges, this research explored the use of a modern deep learning-based object detection model, YOLOv11, for detecting multiple land-cover and infrastructure elements in satellite images.

The primary goal of this project was to design and implement a multi-class object detection system capable of accurately identifying various features such as buildings, roads, vegetation, forests, and water bodies from high-resolution remote sensing images. The study began with a detailed understanding of the limitations of traditional remote sensing analysis techniques, which rely heavily on manual interpretation or handcrafted features. Although these conventional approaches provide useful insights, they often struggle to generalize across different geographic regions and imaging conditions. With the rapid advancement of deep learning in computer vision, automated detection methods have become increasingly effective in handling complex image analysis tasks.

To achieve the objectives of the project, a deep learning-based detection pipeline was developed using the YOLOv11 architecture. YOLOv11 offers several architectural improvements that enhance object detection performance, including improved feature extraction, efficient feature fusion mechanisms, and better handling of objects at

multiple scales. These capabilities make the model particularly suitable for analyzing satellite imagery, where objects may vary significantly in size and orientation.

A key component of the project involved preparing a suitable dataset for training and evaluating the detection model. A custom dataset containing various urban and environmental features was curated and pre-processed to ensure compatibility with the YOLO framework. Data augmentation techniques were applied to increase dataset diversity and improve the model's ability to generalize across different imaging conditions. Additionally, transfer learning techniques were utilized to leverage knowledge from pre-trained models, enabling faster convergence and improved performance during training.

The trained model was evaluated using standard performance metrics such as precision, recall, and detection accuracy. The results demonstrated that the proposed YOLOv11-based framework was capable of accurately detecting multiple object categories in high-resolution remote sensing imagery. The model showed strong performance in identifying different land-cover elements while maintaining efficient detection speed. These results indicate that modern deep learning architectures can significantly improve the automation and accuracy of geospatial image analysis.

The objectives defined at the beginning of the project were successfully achieved. A functional multi-class object detection model was developed and implemented using YOLOv11. The project successfully demonstrated the feasibility of applying advanced deep learning techniques to remote sensing imagery for automated object detection. The dataset preparation, model training, and evaluation processes were completed effectively, and the obtained results highlight the potential of the proposed approach for real-world geospatial applications.

One of the key outcomes of this research is the demonstration that YOLOv11 can effectively address several challenges associated with remote sensing image analysis. The model's ability to detect objects at multiple scales and handle complex spatial patterns contributes to improved detection performance compared to many traditional approaches. Furthermore, the use of data augmentation and transfer learning significantly enhanced the robustness and generalization capability of the system.

In addition to improving detection accuracy, the proposed framework provides a scalable solution that can be extended to larger datasets and additional object categories. The system has the potential to support a wide range of applications, including urban development monitoring, environmental conservation, infrastructure planning, and disaster response. By automating the detection process, the framework reduces the need for manual analysis and enables faster processing of large satellite image datasets.

In conclusion, this project successfully demonstrates the potential of deep learning–based object detection techniques for remote sensing image analysis. The YOLOv11-based framework developed in this study provides an effective approach for identifying multiple urban and environmental features in satellite imagery. While certain challenges such as detecting extremely small objects or handling highly complex scenes may still remain, the results show that the proposed approach represents a significant step toward more accurate and efficient geospatial analysis systems.

Overall, the findings of this research highlight the importance of integrating modern artificial intelligence techniques with remote sensing technologies. Such integration can play a vital role in improving our ability to monitor the Earth's surface, manage natural resources, and support data-driven decision-making for sustainable development.

5.2 Future Scope

The rapid advancement of remote sensing technologies and deep learning techniques opens numerous possibilities for further improving and extending the proposed object detection framework. Although the current project demonstrates the effectiveness of a YOLOv11-based model for detecting multiple land-cover and infrastructure elements in high-resolution satellite imagery, there are several directions in which the system can be enhanced to increase its accuracy, scalability, and practical applicability. Future research can focus on improving model performance, expanding the dataset, integrating advanced technologies, and deploying the system in real-world environments.

5.2.1 Improvements in Model Performance

One of the primary areas for future work involves improving the detection accuracy of the proposed model, particularly for small and densely distributed objects. Remote sensing images often contain objects that appear at very small scales, such as vehicles, small buildings, or narrow roads. Although YOLOv11 provides improved multi-scale detection capabilities, additional enhancements such as advanced attention mechanisms, improved feature pyramid networks, or transformer-based architectures could further enhance the model's ability to capture fine-grained spatial details.

Another potential improvement involves optimizing the training process through the use of larger and more diverse datasets. Increasing the volume and variety of training data can significantly enhance the model's ability to generalize across different geographical regions, imaging conditions, and environmental variations. Future work may also explore semi-supervised or self-supervised learning techniques that allow models to learn from partially labelled or unlabelled remote sensing data, reducing the need for extensive manual annotation.

Hyperparameter optimization and advanced training strategies could also be implemented to further improve model performance. Techniques such as automated neural architecture search, adaptive learning rate scheduling, and improved loss functions may lead to more efficient training and higher detection accuracy.

5.2.2 Extensions to the Project

The current project focuses primarily on multi-class object detection in static satellite imagery. However, the framework can be extended to support additional tasks and functionalities. One possible extension is the integration of semantic segmentation or instance segmentation techniques to provide more detailed information about object boundaries and spatial distribution. This would allow the system not only to detect objects but also to precisely delineate their shapes and areas.

Another potential extension involves incorporating multi-temporal satellite imagery to analyse changes over time. By comparing images captured at different time intervals, the system could be used to monitor urban expansion, deforestation, land-use changes, and environmental degradation. Such temporal analysis would greatly enhance the usefulness of the framework for long-term environmental monitoring and planning.

The project could also be extended to include additional object categories, such as vehicles, agricultural fields, bridges, and industrial infrastructure. Expanding the range of detectable objects would increase the system's versatility and make it applicable to a wider range of geospatial analysis tasks.

5.2.3 Real-World Deployment

For real-world applications, the proposed framework can be integrated into large-scale geospatial information systems or cloud-based remote sensing platforms. Deploying the model on cloud infrastructure would allow it to process large volumes of satellite imagery efficiently and provide real-time or near-real-time analysis. This would be particularly beneficial for applications such as disaster response, where rapid assessment of affected areas is critical.

The system could also be integrated with geographic information systems (GIS) to provide interactive visualization and analysis tools for decision-makers. By combining object detection results with geographic data layers, users could easily identify patterns, trends, and spatial relationships between different environmental and infrastructural features.

Another potential deployment scenario involves the use of edge computing technologies. In such cases, lightweight versions of the detection model could be deployed on drones or satellite ground stations to perform on-site image analysis. This approach would reduce the need for transmitting large image datasets to centralized servers and enable faster decision-making in time-sensitive situations.

In summary, the proposed YOLOv11-based object detection framework provides a strong foundation for automated remote sensing image analysis. With further improvements in model design, dataset expansion, system integration, and advanced AI techniques, the project has the potential to evolve into a powerful tool for large-scale geospatial monitoring, environmental management, and infrastructure planning.

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APPENDIX 1

A. Detection Results on Satellite Images

These figures show the output of the trained object detection model on sample satellite images. The model successfully identifies and labels objects such as buildings, roads, and trees with bounding boxes and confidence scores.

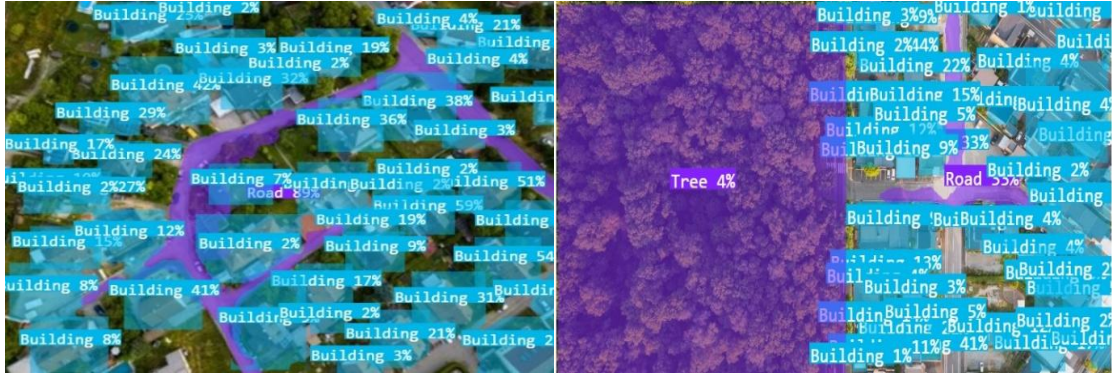


Fig A.1: Detection Results on Satellite Image

B. Model Performance Graph

The graph illustrates the variation of performance metrics, namely **mean Average Precision (mAP)** and **mAP@0.5:0.95**, across training epochs. Both metrics show a consistent upward trend, indicating that the model improves its detection accuracy over time. The stabilization of the curves toward later epochs suggests that the model has achieved convergence with reliable performance.

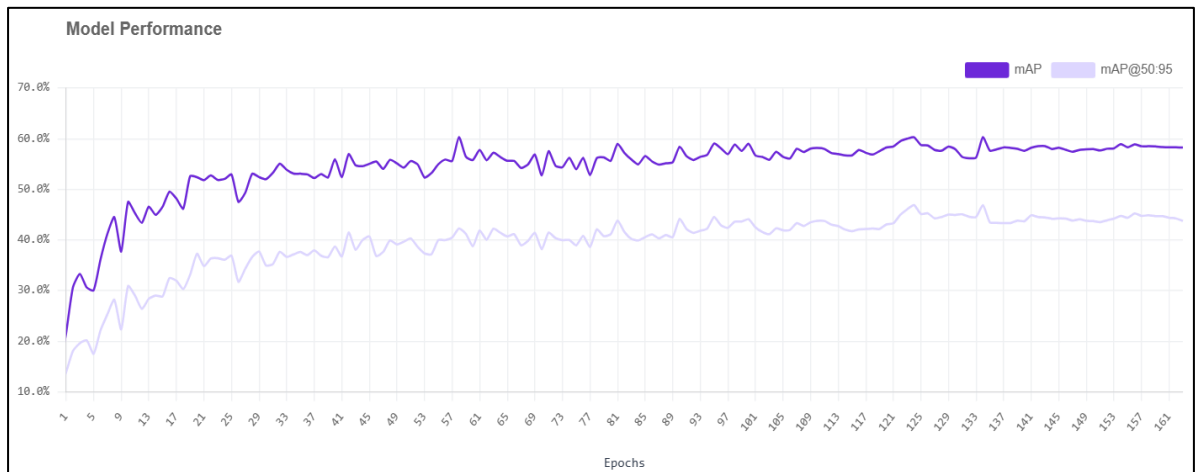


Fig A.2: Model Performance over Training Epochs

C. Training Loss Graphs

- **Class Loss:** Represents the error in class prediction. The decreasing trend shows improved classification accuracy.
- **Object Loss:** Indicates how well the model detects object presence. The gradual reduction signifies better object detection capability.
- **Box Loss:** Measures bounding box prediction error. The decline reflects improved localization accuracy.

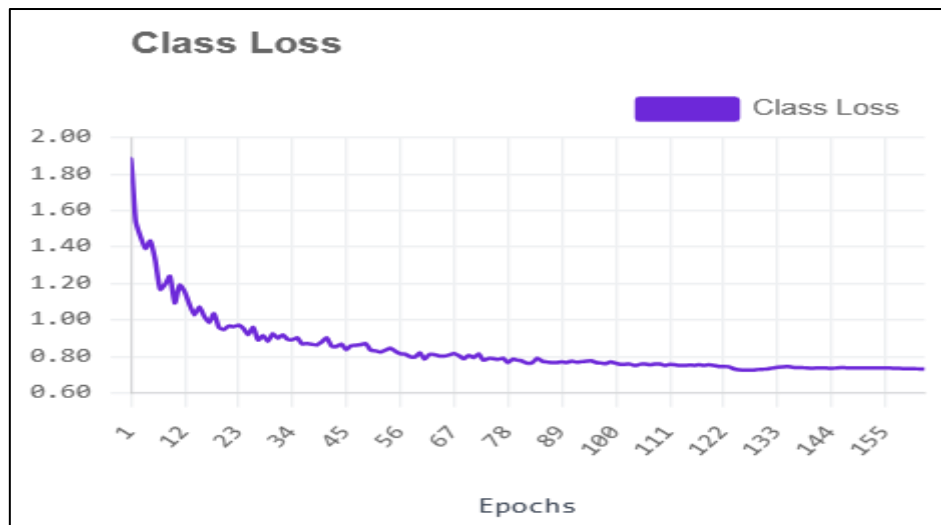


Fig A.3: Class Loss

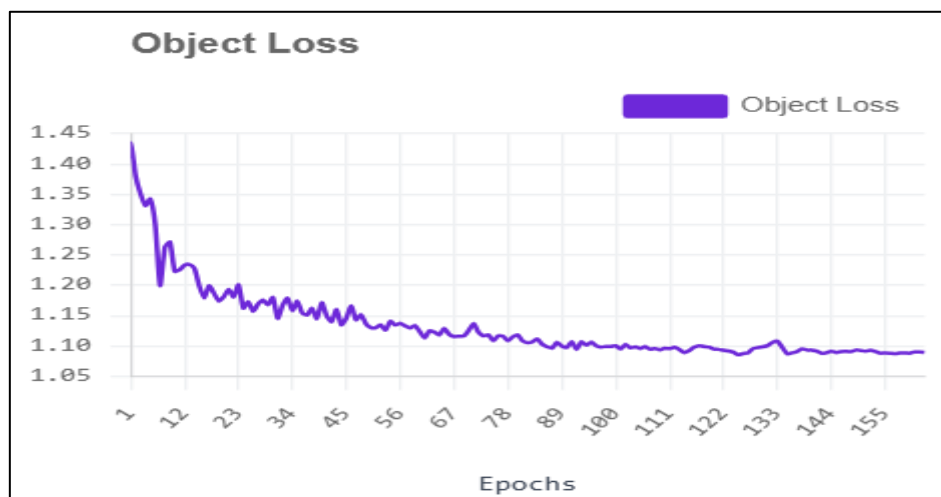


Fig A.4: Object Loss

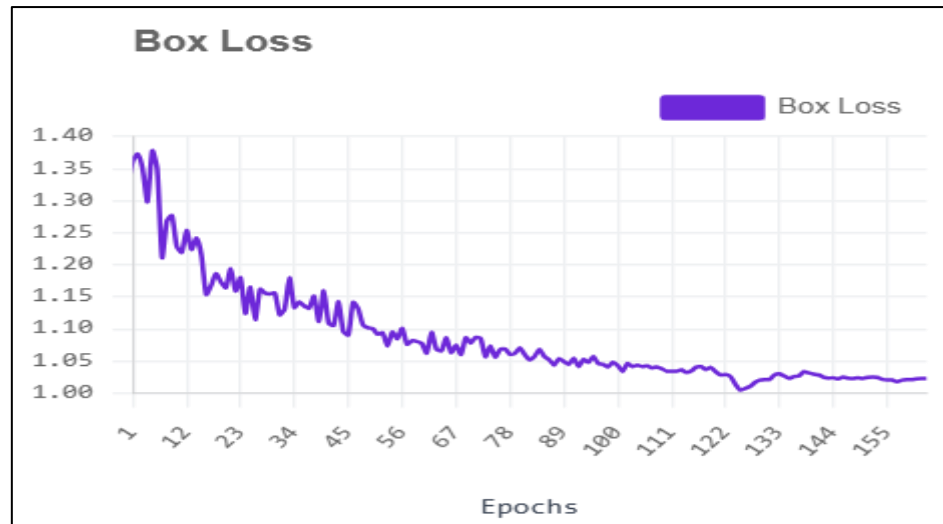


Fig A.5: Box Loss

RESEARCH PAPER

Urban Mapping with YOLOv11

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Abstract—The new trends of satellite and aerial imaging have significantly contributed to the importance of remote sensing in environmental monitoring, disaster response, and urban planning. Despite these developments, problems such as similarity between classes, complex scene structures, and the scarcity of annotated data still affect the accuracy of object detection in high resolution satellite images. Our study presents a novel object detection model grounded in one of the existing deep learning networks, the YOLOv11 algorithm, which provides us with a high level of accuracy and good localization in a challenging environment[1]. Although its predecessors, such as CNN, Faster R-CNN, and variations of YOLO (v5-v8), have been explored and achieved significant results, our proposed plan includes architecture and dynamic data enhancement methods. Our model has been trained and tested on a highly curated dataset in an eclectic combination of natural and urban settings. Our project has been experimentally studied and demonstrated that the YOLOv11 model is better generalized and performs better in determining various urban and environmental aspects even in densely populated areas. The result demonstrates that our proposed model is a reliable and scalable solution for important applications, such as urban development and emergency management response.

I. INTRODUCTION

Technology has advanced to the point where we can remotely analyze satellite images and detect various land covers, such as roads, trees, and buildings, and help cover various areas of application, such as urban planning, environmental monitoring, and disaster management. This is a great milestone, but there is still an issue of object detection in remote environments such as enormous picture sizes, object scales, and complicated backgrounds. Traditional detection designs, including those based on hand-made characteristics or rule based classification are not necessarily capable of generalizing across scenes, are extremely changeable, and cannot scale.

Object Detection Models like Faster Region CNN (R-CNN), You Only Look Once (YOLO), and Retina Net help identify objects in satellite images and use various methods such as region proposal, anchor boxes, and feature pyramid networks

to make object detection better in pictures. Although deep learning has proven to be extremely promising in object recognition and identification of images, this has not been adopted due to the complexity of the image features and the lack of extensive training data. Therefore, additional studies are required to solve the problem of object detection and recognition in the image and to create stronger and more precise deep learning models that can be applied to specific tasks.

Our study uses YOLOv11, a new object detection framework, to overcome the challenges, which are adoptive anchor-free detection, deeper cross-scale attention modules, and better feature fusion. In contrast to the previous research designs that used to identify one object category at a time, our proposed work implements a powerful multi-class detection pipeline that can detect and classify a set of various urban and environmental objects in parallel on top of high-resolution images. The proposed model, based on the implementation of a modified, augmented data set and transfer learning techniques, has better generalization, accuracy, and recall, making YOLOv11 a scalable and robust system to use in real-world geospatial analysis and decision making.

II. LITERATURE REVIEW

A notable application of YOLOv5 is demonstrated in a 2024 study by Wang et al., titled "Multi-class Object Detection in Urban Scenes Based on Deep Learning". The model used in this research utilized YOLOv5s, trained on the ISPRS Potsdam dataset, which provided high-resolution aerial imagery with detailed annotations for various urban elements. The model successfully identified buildings, vehicles, and trees simultaneously. Moreover, it performed much better as compared to earlier objection detection methodologies, such as SSD and Faster R-CNN. The researchers found that YOLOv5s provides a good balance between speed, accuracy, and the size of the model, making it a suitable option for use in smart city projects.

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